

Why the Brain Folding Metaphor Fits Perfectly The Universe as a Folded Brain

Self-Similarity, Surface Maximization, and
Information

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Abstract

This paper examines the parallel between brain folding (cortical gyri) and the 4D torsional structure of T0 theory. The metaphor can be stated mathematically at several points; where it goes beyond a structural analogy, the text says so. Both systems solve the same fundamental problem: **How does one pack maximum surface area/information into minimum volume without singularities?** The analysis describes nine parallels: (1) **Fractal self-similarity** across many scales. (2) **Surface maximization** with volume minimization. (3) **Deep furrows = high density**: Sulci $\leftrightarrow$ mass concentrations. (4) **Singularity avoidance** through minimum curvature radius. (5) **Static structure, dynamic flows**: Material static, information dynamic. (6) **Hierarchical information processing** across levels. (7) **Topological invariants**: Genus = 1 for both. (8) **Energy efficiency** through geometric optimization. (9) **Asymmetry as function**: Left vs. right hemisphere $\leftrightarrow$ cosmic dipoles. The parallels suggest that both systems use a similar geometric solution for information storage and processing; they are to be read as a structural analogy, not as proof of a common mechanism.

Note on versions. This English version of the document is more extensive than the German one. Earlier revisions shortened or changed parts of the German version. Where the two differ, this more extensive English version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. Here this mainly concerns the cosmological statements; the current corpus deliberately sets the cosmic sector aside, since the Standard Model does not derive it either (P39).

.1 Introduction: The Image

The Metaphor

In FFGF/T0 theory, the universe is described as:

"A huge, fractally folded brain"

where the ****deep folds**** (sulci) correspond to regions of highest mass and energy density.

Why is this metaphor fitting?

Central Observation

The human brain and the universe in T0 theory solve ****the same fundamental optimization problem****:

How does one maximize surface area (information, density) in minimum volume without creating singularities (collapse)?

The answer in both cases: **fractal folding**.

.2 Nine Parallels

Parallel 1: Fractal Self-Similarity

Brain

The human cortex shows fractal structure:

- **Large furrows** (primary sulci): 1–2 cm deep
- **Medium convolutions** (secondary sulci): 0.5–1 cm
- **Small folds** (tertiary sulci): 0.1–0.5 cm
- **Microcolumns**: 30–50 μm

Each large fold contains smaller folds following the same principle.

Fractal dimension of cortex: $D_{\text{cortex}} \approx 2.7 - 2.8$

T0 Universe

The torus structure scales self-similarly over ****60+ orders of magnitude****:

Scale	R (major radius)	System
Sub-Planck	$\sim 10^{-39}$ m	Fundamental granulation
Particles	$\sim 10^{-15}$ m	Protons, leptons
Atoms	$\sim 10^{-10}$ m	Electron shells
Planets	$\sim 10^6$ m	Magnetic field torus
Stars	$\sim 10^9$ m	Convection currents
Galaxies	$\sim 10^{20}$ m	Spiral arms
Cosmic web	$\sim 10^{24}$ m	Filaments

Table 1: Self-similar torus structures across scales

Fractal dimension: $D_f = 3 - \xi \approx 2.9998666$

First Parallel

Both systems show ****fractal self-similarity****: Each large structure contains smaller versions following the same geometric principle.

Mathematically: Both fractal dimensions lie between 2 and 3, clearly below 3 for the cortex and very close to 3 for the T0 universe.

- Cortex: $D \approx 2.75$
- Universe: $D \approx 2.9998666$

Parallel 2: Surface Maximization

Brain

Problem: How to pack ~ 16 billion neurons into a skull of ~ 1.3 liters?

Solution: Folding increases surface area.

$$\text{Smooth sphere} \rightarrow A = 4\pi r^2 \approx 600 \text{ cm}^2 \quad (1)$$

$$\text{Folded cortex} \rightarrow A \approx 2400 \text{ cm}^2 \quad (2)$$

Factor 4 more surface area through folding at same volume.

T0 Universe

Problem: How to pack maximum energy density into minimum space without singularities?

Solution: Torus folding.

For a torus:

$$\text{Surface area : } A = 4\pi^2 Rr \quad (3)$$

$$\text{Volume : } V = 2\pi^2 Rr^2 \quad (4)$$

$$\text{Ratio : } \frac{A}{V} = \frac{2}{r} \quad (5)$$

The smaller r (tube radius), the **greater the surface area per volume**.

Limit: $r_{\min} \approx 21\ell_p$ prevents singularity.

Second Parallel

Both systems maximize surface area at minimum volume:

- **Brain:** Maximum neuronal surface area
- **Universe:** Maximum energy density surface area

Both avoid singularities:

- Cortex: Minimum sulcus depth ~ 1 mm (blood supply)
- Universe: Minimum radius $r_{\min} = 21\ell_p$

Parallel 3: Deep Furrows = High Density

Brain

The **deepest sulci** (furrows) of the brain contain the **densest neuronal connections**:

- **Lateral fissure** (Sylvian fissure): Separation frontal/temporal lobes
 - $\rightarrow$ Language centers (Broca, Wernicke)
 - $\rightarrow$ high cognitive density
- **Central sulcus:** Motor/sensory cortex
 - $\rightarrow$ Direct body control
 - $\rightarrow$ Maximum information density

Principle: Deep folds $\leftrightarrow$ high functional importance

T0 Universe

The **deepest folds** of the torus geometry (regions with negative Gaussian curvature) correspond to **highest mass densities**:

Gaussian curvature of torus:

$$K(\theta) = \frac{\cos \theta}{r(R + r \cos \theta)} \quad (6)$$

Outside ($\theta \approx \pi$): $K < 0 \rightarrow$ **Negative curvature**

Here T0 theory places **[S]**:

- Galaxy cores
- Supermassive black holes
- Supercluster nodes

- Filament intersection points

Third Parallel

Deep furrows = High density

Brain	Universe (T0)
Deepest sulci	Negative curvature ($K < 0$)
↓	↓
Densest neuronal connections	Highest mass density
↓	↓
Maximum information	Maximum energy

Parallel 4: Singularity Avoidance

Brain

The cortex cannot fold ****arbitrarily deep****:

Limitations:

1. **Blood supply**: Deep sulci need capillaries
2. **Mechanical stability**: Too thin walls collapse
3. **Minimum thickness**: $\sim 1.5 - 4$ mm (gray/white matter)
 $\Rightarrow$ Minimum curvature radii prevent "singularities"

T0 Universe

The fractal dimension $D_f = 3 - \xi$ prevents collapse:

In perfect 3D space ($D = 3$): Torus could shrink to $r \rightarrow 0$ (singularity).

With $D_f = 3 - \xi$: Minimum tube radius

$$r_{\min} \propto \frac{\ell_P}{\xi^{1/3}} \approx 21 \times \ell_P \approx 3.4 \times 10^{-34} \text{ m} \quad (7)$$

Meaning: In the T0 description, the fractal structure of space itself prevents singularities.

Fourth Parallel

Both systems avoid singularities through natural minimum curvature radii:

- **Brain**: $r_{\min} \sim 1$ mm (biological)
- **Universe**: $r_{\min} \sim 21\ell_P$ (geometrical)

The folding increases surface area ****without collapsing into singularities****.

Parallel 5: Static + Dynamic

Brain

Structure: Materially ****static****

- Neurons don't move
- Cortex architecture is fixed
- Anatomy remains constant
 Function: Electrically ****dynamic****
- Action potentials propagate
- Synapses fire
- Information flows

T0 Universe

Structure [S]: In this reading the universe is ****static****

- No Big Bang
- No cosmic expansion
- 4D torsion crystal is timeless
 Dynamics: Energy flows are ****dynamic****
- Photons propagate
- Torsion waves travel
- Energy circulates in torus

Redshift [S]: In this reading it arises not from expansion, but from:

$$z \approx \xi \cdot \ln\left(\frac{d}{\ell_P}\right) \quad (8)$$

Fractal energy loss along the folds. These cosmological statements are speculative; the cosmic sector is deliberately set aside in the current corpus (P39).

Fifth Parallel

Static base structure, dynamic flows:

	Brain	Universe (T0)
Material/Structure	Static	Static
Information/Energy	Dynamic	Dynamic
Surface/Space	Folded	Folded (torus)

Parallel 6: Hierarchical Processing

Brain

Neuronal information processing is ****hierarchical****:

1. **Level 1:** Receptors (retina, cochlea)
2. **Level 2:** Primary sensory areas (V1, A1)
3. **Level 3:** Secondary areas (V2, V4)
4. **Level 4:** Association cortex
5. **Level 5:** Prefrontal cortex (executive function)
 Each level extracts more abstract features.

T0 Universe

Torsion structures are nested across scales:

1. **Sub-Planck:** $\Lambda_0 \sim 10^{-39}$ m – Fundamental granulation
2. **Planck:** $\ell_P \sim 10^{-35}$ m – Quantum gravity
3. **Particles:** $\sim 10^{-15}$ m – Protons, leptons
4. **Atoms:** $\sim 10^{-10}$ m – Electron shells
5. **Stars:** $\sim 10^9$ m – Convection torus
6. **Galaxies:** $\sim 10^{20}$ m – Spiral arms
7. **Cosmic:** $\sim 10^{24}$ m – Filament network

Each scale is described as a torus, **embedded in larger tori**.

Sixth Parallel

Hierarchical information processing:

- **Brain:** Neural networks on different abstraction levels
- **Universe:** Nested torus vortices from Planck to Hubble

Both are **fractally layered**.

Parallel 7: Topological Invariance

Brain

Topologically, the cortex can be regarded as a **torus**.

Why?

- Cerebral hemispheres are connected by the **corpus callosum**
- The ventricular system forms a **central hole**
- Genus = 1 (one hole)

Mathematically: In this simplification, the folded cortex can be continuously deformed into a torus.

T0 Universe

The fundamental structure is a **4D torus**:

$$\mathcal{M} = \mathbb{R}^3 \times S^1_{\text{comp}} \quad (9)$$

Properties:

- 3 spatial + 1 compact dimension
- Genus = 1 (one hole)
- Poloidal + toroidal circulation

Seventh Parallel

Both have the same topology: Torus (Genus = 1)

At the level of topology this is more than a metaphor: both have the same genus.

- Cortex: Topologically equivalent to torus
- Universe: Fundamental 4D torus

The topology is **invariant** under folding.

Parallel 8: Energy Efficiency

Brain

The brain is **very energy efficient**:

- Power: ~ 20 Watts
- Operations: $\sim 10^{16}$ synapses/second
- Efficiency: $\sim 10^{-15}$ Joules per operation

Reason: Folding shortens wiring (axons) while keeping connectivity high.

Principle: Minimize

$$E_{\text{total}} = E_{\text{wiring}} + E_{\text{volume}} \quad (10)$$

$\Rightarrow$ Solution: folded surface.

T0 Universe

The torus minimizes energy for given topology:

$$E_{\text{total}} = E_{\text{surface}} + E_{\text{curvature}} + E_{\text{rotation}} \quad (11)$$

Variational calculus shows: For constant flux and angular momentum, the torus is the **most stable form**.

The fractal dimension $D_f = 3 - \xi$ means:

- Energy experiences "resistance" when flowing
- Torus is the path of **least resistance**

Eighth Parallel

Both systems optimize energy:

- **Brain:** Minimum wiring, maximum function
- **Universe:** Minimum energy, maximum stability

The folding is the **solution to a variational problem**.

Parallel 9: Asymmetry as Function

Brain

The brain is **asymmetric**:

- **Left hemisphere:** Language, logic, sequential
 - **Right hemisphere:** Spatial, holistic, parallel
- This asymmetry is **functional**, not a defect.
- Folding pattern:** Left and right different
- Left Sylvian fissure: Deeper (language center)
 - Right parietal lobe: Larger (spatiality)

T0 Universe

The universe shows **asymmetries** that T0 theory interprets as intrinsic **[S]**:

- **CMB dipole**: Preference direction in cosmic microwave background
- **Cosmic flows**: Large-scale movements
- **Two-dipole model**: Fundamental asymmetry of the "global fold"

In T0 theory this asymmetry is interpreted as structural rather than as a disturbance (speculative, cosmic sector, P39).

It arises from **pentagonal symmetry breaking** by the golden ratio φ :

$$\xi = \frac{4}{30000} \quad \text{with factor } 5\varphi \text{ in the structure} \quad (12)$$

Ninth Parallel

Asymmetry is functional:

Brain	Universe (T0)
Left vs. right hemisphere	CMB dipole, cosmic flows
Functional specialization	Global asymmetry of fold
Emerges from development	Emerges from φ -breaking [S]

.3 How far does the metaphor carry?

Mathematical Precision

Some of the parallels can be stated **quantitatively**:

Property	Brain	Universe (T0)
Fractal dimension	$D \approx 2.75$	$D_f = 3 - \xi \approx 2.9998666$
Topological genus	1 (torus)	1 (4D torus)
Surface gain	$\times 4$	$\propto 1/r_{\min}$
Minimum radius	$\sim 1 \text{ mm}$	$21\ell_p$
Hierarchy levels	$\sim 5 - 6$	> 60

Table 2: Quantitative parallels

Universal Optimization Principle

Both solve a comparable problem through **a similar geometric strategy**:

Maximize $\frac{\text{Surface (Information)}}{\text{Volume (Space)}}$
under the constraint:
No singularities.

Information is Geometry

The interpretation that follows:

Information = Geometry

In both systems, information is geometrically encoded.

Brain:

- Neuronal information $\leftrightarrow$ folding structure
- More surface = more synapses = more information

Universe:

- Physical information $\leftrightarrow$ torsion structure
- More windings = more energy = more information

The metaphor suggests: **geometry carries information**.

.4 The Narrative Power

Why brain instead of other metaphors?

There are many folded systems (paper, fabric, intestine, ...). Why is the **brain** so fitting?

Why brain?

1. Consciousness and cosmos:

The brain is the most complex known object in the universe. The metaphor suggests: The universe itself might have a form of "consciousness" – not in an anthropomorphic sense, but as a **self-organizing information system**.

2. Micro-macro unity:

The smallest conscious system (brain, ~ 1 kg) and the largest system (universe, $\sim 10^{53}$ kg) follow, in this reading, **similar geometric principles**.

This corresponds to a core statement of T0 theory: **self-similarity over 60 orders of magnitude**.

3. Emergence and complexity:

From simple folding rules (torus geometry) emerges great complexity:

- Brain: ~ 86 billion neurons, $\sim 10^{14}$ synapses
- Universe: $\sim 10^{80}$ particles, cosmic web

Both are **more than the sum of their parts**.

The holographic principle

The brain folding metaphor connects with the **holographic principle**:

Holography

Holographic principle: The information of a volume is encoded on its surface.

Brain: The ~ 2 mm thin cortex ****surface**** contains all cognitive information – the underlying volume (white matter) serves mainly as wiring.

Universe (T0): The torsion ****surface**** (4D hypersurface) encodes all physical information – the "volume" is emergent **[S]**.

Folding increases surface $\Rightarrow$ increases information capacity.

.5 Summary: Nine Parallels

No.	Parallel	Brain	Universe (T0)
1	Fractal self-similarity	Sulci at all scales	Torus structures 60+ orders of magnitude
2	Surface maximization	$\times 4$ through folding	$\propto 1/r_{\min}$
3	Deep furrows = density	Neuronal density in sulci	Mass density at $K < 0$
4	Singularity avoidance	$r_{\min} \sim 1$ mm	$r_{\min} = 21\ell_p$
5	Static + dynamic	Material static, electrical dynamic	Structure static, energy dynamic
6	Hierarchical processing	5-6 cortical levels	7+ scale levels
7	Topology: torus	Genus = 1	4D torus
8	Energy efficiency	Minimum wiring	Minimum energy
9	Asymmetry as function	Left vs. right	CMB dipole [S]

Table 3: The nine parallels

.6 Conclusion

Why does the metaphor fit?

The brain folding metaphor fits for the following reasons:

1. Mathematical common ground: Both have fractal dimensions in the range $D \approx 2.7 - 3.0$ and can be described by torus topology (genus = 1).

2. Same optimization problem: Both maximize surface area/information at minimum volume without singularities.

3. Self-similarity: Both show fractal hierarchy across many scales.

4. Information = geometry: Both encode information in folded surface.

5. Narrative depth: The metaphor connects the smallest conscious system (brain) with the largest system (universe) and raises the question whether ****consciousness and cosmos are geometrically related**** **[S]**.

The parallels point to a ****common geometric solution**** for information storage and processing; whether a common principle lies behind it remains an open question.

Summary interpretation

Interpretation

The universe doesn't think like a brain –

The brain is folded like the universe.

Both can be described with the same geometric logic:

$$\max \left(\frac{\text{Surface}}{\text{Volume}} \right) \text{ with } r \geq r_{\min} \quad (13)$$

The solution in both cases: **fractal folding in torus topology.**